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Tuning Structural and Optical Properties of Thiolate-Protected Silver Clusters by Formation of a Silver Core with Confined Electrons

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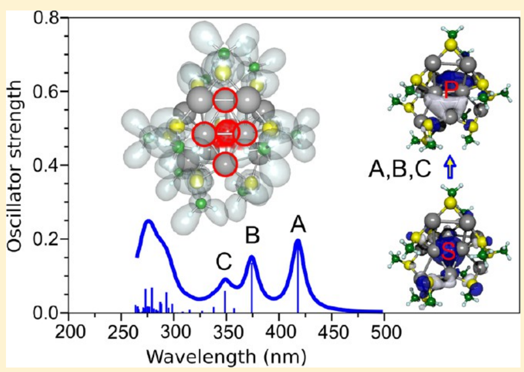
S *Supporting Information*

ABSTRACT: We present a systematic theoretical investigation of the structural and optical properties of thiolate-protected silver clusters with the goal to design species exhibiting strong absorption and fluorescence in the UV−vis spectral range. We show that the optical properties can be tuned by creating systems with different counts of confined electrons within the cluster core. We consider liganded silver complexes with n silver atoms (Ag_n) and x ligands (L_x) in anionic complexes $[Ag_nL_x]^-$ with L = SCH_3. Variation of the composition ratio gives rise to systems with (i) zero confined electrons for $x = n + 1$, (ii) two confined electrons for $x = n - 1$, and (iii) four confined electrons for $x = n - 3$. We show that the number of confined electrons within the cluster core and the geometric structure of the latter are responsible for the spectral patterns, giving rise to intense absorption transitions and fluorescence in the visible or even infrared range. Our results open a perspective for the rational design of stable ligand-protected silver cluster chromophores that might find numerous applications in the field of biosensing.



INTRODUCTION

The unique electronic and optical properties of noble metal clusters in the nonscalable size regime characterized by discrete electronic states and structure-dependent optical properties,[1−3] which differ significantly from those observed in bulk materials, have attracted broad interest. In particular, the presence of strong visible or near-infrared emission[4−7] has drawn the attention of researchers due to possible applications in the fields of biosensing, single-molecule spectroscopy, and optoelectronics. In this context, the stabilization of noble metal clusters and their protection are of essential importance. In fact, the preparation of gold−thiolate nanoclusters has been very successful already more than a decade ago.[4] In the meantime, the synthesis of almost monodispersed thiolated nanoclusters has been experimentally realized.[8−14] Ligand-protected clusters with 25 gold atoms have been particularly intensively investigated.[15−17] It has been found that the pronounced high stability of these species arises due to the formation of an almost icosahedral Au_{13} cluster core on which surface-protecting V-shaped −S−Au−S−Au−S units are distributed in an approximate octahedral arrangement.[18−21] The influence of various types of thiolates on the stability of ligand-protected gold and silver clusters has been theoretically investigated.[22,23] In particular, the enhanced stability of the highly symmetric $Au_{144}(SR)_{60}$ cluster has been found to be of geometric origin, while the electronic properties are well-described using the "superatom complex model".[24] In addition to determining the structural properties and stability, the nature of ligands also seems to strongly influence the fluorescence yield of ligand-protected gold clusters,[25] which is of large interest in the context of applications exploiting the unique absorption and emission properties of such systems.

In contrast to gold, the preparation of silver clusters protected by glutathione ligands has been reported recently, indicating that the growth mechanism might be different[26] from the one for the thiolate-protected gold species. Therefore, also the structural and optical properties of thiolated silver clusters may exhibit novel features. Specifically, due to the considerably larger relativistic effects in gold, resulting in the small s−d energy gap, the d-electrons contribute substantially to the low-lying excitations and thus influence the spectroscopic patterns. In contrast, in the case of silver, the s−d energy gap is much

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larger, and s−p type excitations lead to strongly localized intense absorption, which is preferable in the context of biosensing applications (compare ref 27). This means that, although structural properties of silver and gold glutathione oligomers are similar, their optical properties in the UV range differ substantially[28] even in systems containing a single metal atom. Therefore, the theoretical study of structural and optical properties of ligand-protected silver clusters is of significant importance because it might allow for guiding the experimental preparation and characterization of these systems. This can be achieved by systematic variation of the system composition (number of silver atoms versus number of ligands), leading to the prediction of species with desired absorption and fluorescence properties. In particular, because strong absorption in the UV−vis range is typically present in systems with a certain number of confined electrons, the fundamental question can be raised at which compositions ligand-protected silver clusters will exhibit both high stability as well as intense absorption and fluorescence. In fact, gas-phase reactivity performed in an ion trap has been recently used to produce small anionic silver clusters (Ag_3, Ag_4) protected with thioglycolic acid. Oxidation−reduction reactions lead to the formation of the predicted systems with two confined electrons within the cluster subunit, which show characteristic absorption patterns in the visible range.[29] Moreover, synthesis and characterization of optical properties of $Ag_{31}(SG)_{19}$ and $Ag_{15}(SG)_{11}$ have provided evidence that $Ag_{15}(SG)_{11}$ with four confined electrons within the cluster core exhibits fluorescence with a quantum yield of 2.2%.[30] In addition, the crystal structure of a luminescent thiolated Ag nanocluster with an octahedral Ag_6^{4+} core has been recently experimentally synthesized and characterized. Its structure resolved by single-crystal X-ray analysis shows substantial differences with respect to model structures of thiolate-capped Au clusters.[31]

In this contribution, we investigate theoretically the optical properties of anionic thiolate-protected silver clusters with the aim to design systems exhibiting strong absorption and emission in the visible range. For this purpose, we systematically vary the chemical composition in order to control the number of confined electrons within the cluster core formed of those Ag atoms that are not directly bound to ligands. As a prototype ligand, we have chosen the SCH_3 group. Because the sulfur present in thiols can be considered as an electron acceptor, each pair of Ag and S atoms formally consists of Ag^+ and S^-; thus, the excess of Ag atoms with respect to the number of ligands containing sulfur determines the number of electrons that can be confined within the cluster core. For the anionic species, one additional electron has to be added to the electron counting. Therefore, we consider systems with n silver atoms (Ag_n) and x ligands (L_x) in anionic complexes $[Ag_nL_x]^-$, which according to the electron count can give rise (i) to zero confined electrons for $x = n + 1$, (ii) to two confined electrons for $x = n - 1$, and (iii) to four confined electrons for $x = n - 3$. We have chosen to study anionic complexes because, in general, charged species can be investigated experimentally using mass spectrometry combined with action spectroscopy, as in our previous joint theoretical and experimental studies of different classes of hybrid systems containing metal clusters.[32−36] Particular attention will be paid to the formation of cluster cores containing two or four confined electrons and to the influence of the cluster core on the optical absorption and fluorescence patterns of these complexes. The classification according to the number of confined electrons within the cluster core will allow us to propose ligand-protected silver species with optical properties suitable for applications such as fluorescence labeling.

After a brief computational section, the results and discussion are presented for different classes of complexes with 0, 2, and 4 confined electrons, and finally, conclusions are given.

■ COMPUTATIONAL

Density functional theory (DFT) has been used to determine the structural properties of ligand-protected silver complexes, employing the PBE functional[37] and the relativistic effective core potential (RECP) of the Stuttgart group for the silver atoms.[38] The TZVP atomic orbital basis set was used for all atoms.[39] An extensive search for structures was performed using simulated annealing coupled to molecular dynamics (MD) simulations in the frame of the semiempirical AM1 method[40] with parameters for Ag atoms from ref 41. The structures identified in this way were subsequently reoptimized at the DFT level using gradient minimization techniques, and stationary points were characterized by calculating the harmonic vibrational frequencies. Notice that low-energy structures obtained by DFT differ substantially from those reached by the AM1 method, in particular, for larger complexes. Therefore, despite an extensive search for the lowest-energy structures, finding the global DFT minimum is not fully guaranteed. The electron localization functions (ELFs)[42] have been calculated in order to analyze the nature of bonding and the distribution of confined electrons within the cluster cores. The absorption spectra were computed in the frame of time-dependent density functional theory (TDDFT) employing the long-range corrected hybrid CAM-B3LYP functional,[43] which provides a more reliable description of charge-transfer transitions, as shown by comparing the theoretical and experimental results in our previous work.[29,30] For the calculation of fluorescence spectra, the structures have been optimized in the first excited state using TDDFT/CAM-B3LYP, and harmonic vibrational frequencies as well as normal modes were determined. Subsequently, a harmonic Wigner distribution was generated at 0 K around the excited-state minima, and the excitation energies and oscillator strengths were determined for 100 structures sampled from this distribution. Finally, the envelope of the fluorescence spectrum of the ensemble has been obtained by convoluting the individual transitions by a Lorentzian function.

■ RESULTS AND DISCUSSION

Complexes without Confined Electrons. The structural properties of anionic ligand-protected silver clusters with n silver atoms and $x = n + 1$ ligands, which do not contain a cluster core and therefore no confined electrons, are characterized by two classes of structures built from either chains[21] or rings of $-SCH_3-Ag-SCH_3$ subunits. Analogous complexes have been previously found for thiolate-protected gold oligomers.[44] In particular, we focus here on complexes with n = 3−5 and 8 silver atoms in order to investigate the influence of structures and size on optical absorption patterns. Both the chains and rings contain strongly directional and rigid S−Ag−S bonds. Because there are no silver−silver bonds present, the absorption spectra are characterized by many transitions with comparable intensities located in the UV range below 350 nm, as illustrated in Figure 1 for two prototype examples with n = 4 and 8. In the case of the complex with n =

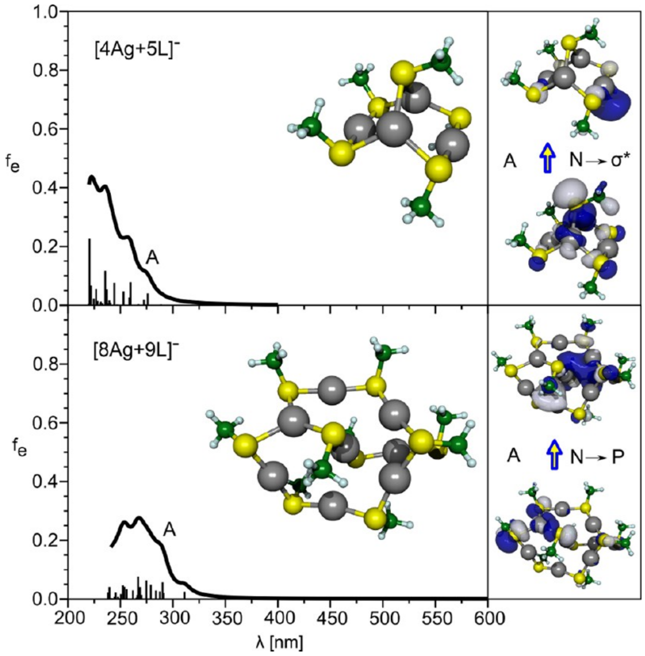

Figure 1. Calculated absorption spectra for $[4Ag+5L]^-$ and $[8Ag+9L]^-$ ($L = SCH_3$) complexes with zero confined electrons. The lowest-energy structures are shown. The spectra have been broadened by a Lorentzian function with a width of 15 nm. Leading excitations and involved orbitals are drawn on the right-hand side.

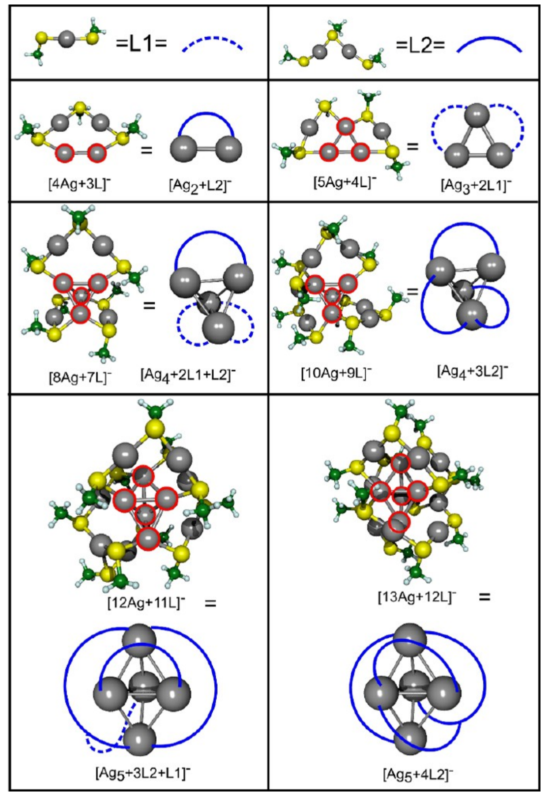

Figure 2. Structures of $[n\text{Ag}+(n-1)\text{L}]^-$ (n = 4, 5, 8, 10, 12, 13) complexes with two confined electrons illustrating the formation of a core (red circles) that is protected by the two types of ligands L1 = SCH_3−Ag−SCH_3 or L2 = SCH_3−Ag−SCH_3−Ag−SCH_3.

4 Ag atoms, a −Ag−S−Ag− ring is formed containing four Ag atoms and four SCH_3 ligands. The ring structure is bridged by the fifth SCH_3 ligand, which connects two Ag atoms at the opposite diagonal positions within the ring structure. In the system with $n = 8$ Ag atoms, a six-membered ring consisting of alternating −Ag−S−Ag− units is bridged by a −S−Ag−S−Ag−S chain. The discussion is focused on the lowest-energy isomers. For an overview of higher-energy isomers that are either related to the lowest-energy classes of structures or contain chain-like arrangements of silver atoms, compare Figure S1 (Supporting Information) and ref 22.

For both systems with zero confined electrons, the lowest-lying electronic transitions with weak intensities located between 270 and 290 nm arise due to excitations from the nonbonding p orbitals localized at sulfur atoms to the P-like delocalized virtual orbitals, as shown in the right-hand part of Figure 1. From the analysis of the absorption spectra and the involved transitions, it can be expected that thiolate-protected silver clusters with zero confined electrons will have similar spectroscopic patterns, independent of the system size. The absence of intense transitions in the visible regime due to the absence of silver subunits does not make them good candidates for the application as fluorescent labels.

Complexes with Two Confined Electrons. In the case of anionic ligand-protected silver clusters containing n silver atoms and $x = n - 1$ ligands, two confined electrons are present within the silver cores. In Figure 2, we show the schematic classification of the structures for systems with n = 4, 5, 8, 10, 12, and 13. An intensive structure search revealed that the energetically lowest isomers of all investigated species can be divided into a silver core and a set of staple ligands with the composition S−Ag−S (L1) and S−Ag−S−Ag−S (L2). The Ag−Ag distances of 3.1−3.2 Å within the staple ligand and toward the cluster are significantly longer than the Ag−Ag bond lengths within the cluster core (2.8−2.9 Å), which allows for a distinct separation into cluster core and staple ligands. It should be pointed out that these structural motifs are different than those of large thiolate-protected silver clusters[45] as well as to those of self-assembled monolayers on Ag(111) surfaces.[46]

The two smallest investigated systems, $[3Ag+2L]^-$ and $[4Ag+3L]^-$, show a Ag_2 core protected by an L1 or L2 staple, respectively. In the case of $[5Ag+4L]^-$, the core of Ag_3 is protected by two L1 ligands. Systematic increase of the system size by adding $AgSCH_3$, leading, for example, to $[8Ag+7L]^-$ and $[10Ag+9L]^-$, gives rise to the Ag_4 core, the size of which remains the same until most of the L1 staples are replaced by L2 ligands. Similarly, in the case of $[12Ag+11L]^-$ and $[13Ag+12L]^-$ with a Ag_5 core, the latter one is protected by L2 ligands only. Moreover, staples bind not only to the core but also to the silver atom of another staple, indicating that further growth of the cluster core with two confined electrons is unfavorable. Thus, an octahedral core as it is predicted for the smallest gold cluster containing two confined electrons ($[Au_{12}(SCH_3)_9]^+$) might not be favorable in anionic liganded silver.[47] Notice that for all systems with a silver core larger than Ag_2, at least one core atom is bound to two sulfur atoms from different staples, which is one of the differences in structural properties with respect to thiolate-protected gold clusters.

In the smallest system $[3Ag+2L]^-$, in which a silver dimer is bound to the L1 ligand, the absorption spectrum is characterized by dominant transitions located at ∼390 nm and two additional weaker ones at 320 and 330 nm [cf. Figure 3]. The analysis of the transitions shows that excitations from S-type delocalized orbitals to the three components of P-type

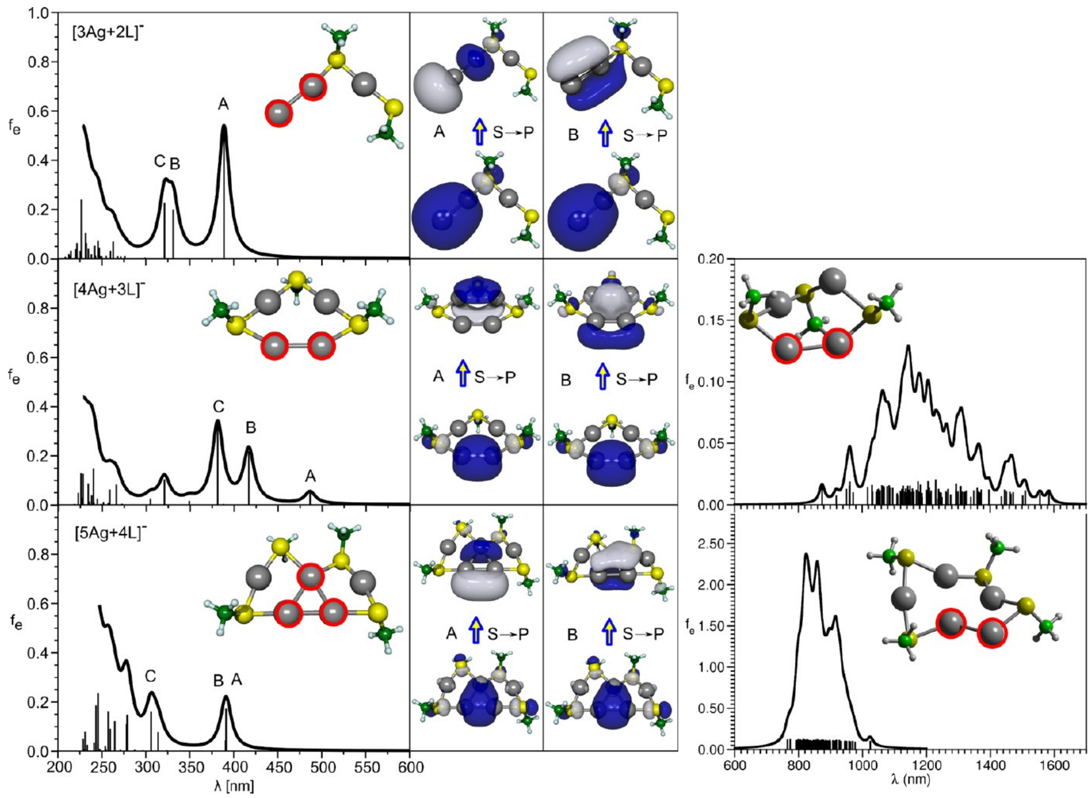

Figure 3. (Left) Calculated absorption spectra for $[n\mathrm{Ag}+(n-1)\mathrm{L}]^-$ complexes (n = 3−5; L = SCH_3) with two confined electrons. The spectra have been broadened by a Lorentzian width of 15 nm. (Middle) Leading excitations and involved orbitals in transitions A, B, and C arising between S and P orbitals (only two of three P components are shown) of the silver core (red circles) containing two and three silver atoms. (Right) Broadened fluorescence spectra obtained from 100 structures sampled from a 0 K harmonic Wigner distribution around the optimized geometry of the S_1 state.

orbitals of the silver dimer are responsible for these intense transitions (cf. right side of Figure 3, where only two P orbital components are shown). For complexes with four and five silver atoms, the Ag_2 core protected by L2 and the Ag_3 core protected by two L1 units are formed, respectively. Due to the presence of two confined electrons, three transitions, A, B, and C, are again present, arising from excitations between the S and P_x, P_y, and P_z orbitals of the silver core. The location of these three characteristic transitions depends on the size of the silver core. In the case of the Ag_2 core, the lowest-energy transition (A) with weak intensity is located at ∼500 nm, while the B and C transitions occur at 450 and 350 nm, respectively. In contrast, the system containing the Ag_3 core gives rise to degenerate transitions A and B at ∼400 nm, while the third transition C is located at ∼300 nm (cf. Figure 3). The structures of two higher-energy isomers for $[4\mathrm{Ag}+3\mathrm{L}]^-$ and $[5\mathrm{Ag}+4\mathrm{L}]^-$ shown in Figure S2 (Supporting Information) contain an unprotected silver atom, and consequently, their absorption patterns differ from those calculated for the lowest-energy isomers.

Because the intense transitions of these smallest liganded clusters with two confined electrons are located in the visible regime, they might be good candidates to exhibit also visible fluorescence. Therefore, we calculated the fluorescence spectra for $[4\mathrm{Ag}+3\mathrm{L}]^-$ and $[5\mathrm{Ag}+4\mathrm{L}]^-$ with Ag_2 and Ag_3 cores, as shown on the right side of Figure 3 together with the structures corresponding to the minima of the lowest excited states (S_1). The geometry relaxation in the excited state of $[\mathrm{Ag}_4+3\mathrm{L}]^-$ leads to a very strong red shift of the maximum of the fluorescence spectrum with respect to the absorption transition, such that the fluorescence maximum occurs at 1100 nm. In contrast, in the case of $[5\mathrm{Ag}+4\mathrm{L}]^-$, the shift of the fluorescence maximum is substantially smaller, giving rise to maximal fluorescence at 850 nm. This indicates that for increasing size of the core and its protection, smaller geometrical relaxation in excited states can be expected.

The question can be raised whether the spectroscopic pattern will change if other ligands are used. For this reason, we replaced SCH_3 by SCH_2COOH in the $[5\mathrm{Ag}+4\mathrm{L}]^-$ complex with two confined electrons. The Ag_3 core remains present in this system, and the spectroscopic pattern characterized by A- and B-type transitions between 370 and 400 nm remains also almost unchanged (cf. Figure S3, Supporting Information). However, the Stokes shift of the fluorescence band might be dependent on the nature of ligands due to their possible effect on geometric relaxation in excited states.

By further increasing the number of silver atoms in complexes with two confined electrons, either a Ag_4 or Ag_5 core is formed, as shown in Figures 4 and 5 for the found lowest-energy isomers (for other isomers see Figure S2, Supporting Information). In both cases, the cluster core is fully protected by the ligands. Again, the three characteristic transitions that are fingerprints of two confined electrons are present in all cases, but the size of the core influences their location. Interestingly enough, in the case of systems with n = 12 and 13 Ag atoms, a core of Ag_5 is formed, and the lowest-energy intense transition is located at around 400 nm, as shown in Figure 5. This means that ligand protection provides a red shift compared to the absorption transitions of the bare charged and neutral clusters.[1,32,33] Such a shift of the optical absorption toward the visible spectral range is a highly desirable property

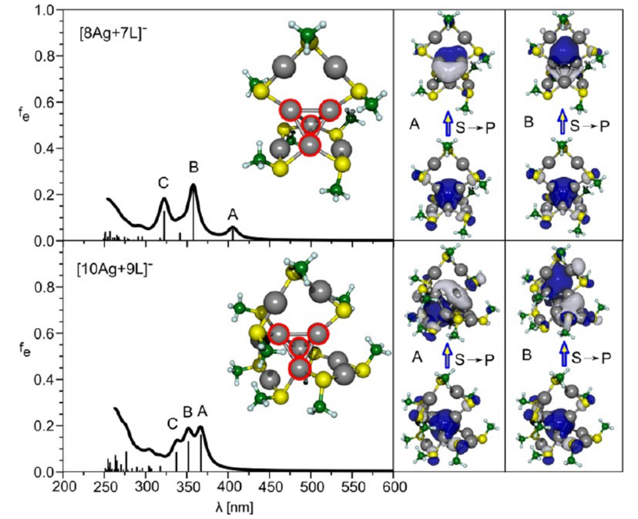

Figure 4. Calculated absorption spectra for $[n\text{Ag}+(n-1)\text{L}]^-$ complexes (n = 8, 10; L = SCH_3) with two confined electrons. The spectra have been broadened by a Lorentzian function with a width of 15 nm. Leading excitations and involved orbitals in transitions A, B, and C arising between S and P orbitals (only two of three P components are shown) of the silver core (red circles) containing four silver atoms are presented on the right-hand side.

in the context of the development of novel superior optical labeling species. Moreover, because all lowest-lying intense transitions correspond to the lowest excited states and are well-separated from other electronic states, these systems might also exhibit strong fluorescence provided that competing non-radiative relaxation processes are of low efficiency. In fact, for $[12\text{Ag}+11\text{L}]^-$, the position of the fluorescence line has been determined to be about 680 nm for the optimized geometry of the first excited state, which is also shown in Figure 5. The cluster core remains present in this geometry, which is reflected by the fact that the fluorescence maximum is red shifted by no more than 250 nm with respect to the corresponding absorption transition (cf. also refs 30 and 48). These results are promising in the context of novel optical fluorescence labels that might be developed based on thiolate-protected silver clusters.

Another open question that we would like to address here theoretically concerns the stability of such ligand-protected species. In order to examine how the stability depends on the size of the system, the incremental binding energy of Ag−SCH_3 subunits to the smallest possible species containing two confined electrons, $[2\text{Ag}+\text{L}]^-$, is shown in Figure 6a. The incremental binding energy increases up to the size of eight silver atoms and reaches a stability plateau with a weak maximum for $[12\text{Ag}+11\text{L}]^-$. Because thiolate ligands are redox-active, it is an open question if the number of confined electrons might be increased by dissociation of two such ligands. In this case, the oxidation of these ligands to a dimer via formation of a disulfide bond would give rise to two excess electrons that could be confined in the cluster core. Therefore, we have determined the dissociation energy for the reaction leading from a system without confined electrons to its counterpart containing two confined electrons by formation of the disulfide ($CH_3S{-}SCH_3$), as shown in Figure 6b. It can be seen that the dissociation energy decreases with increasing system size until reaching a minimum for 12 silver atoms and subsequently increases for larger sizes. It is noteworthy that the dissociation energy for the system containing 12 Ag atoms is

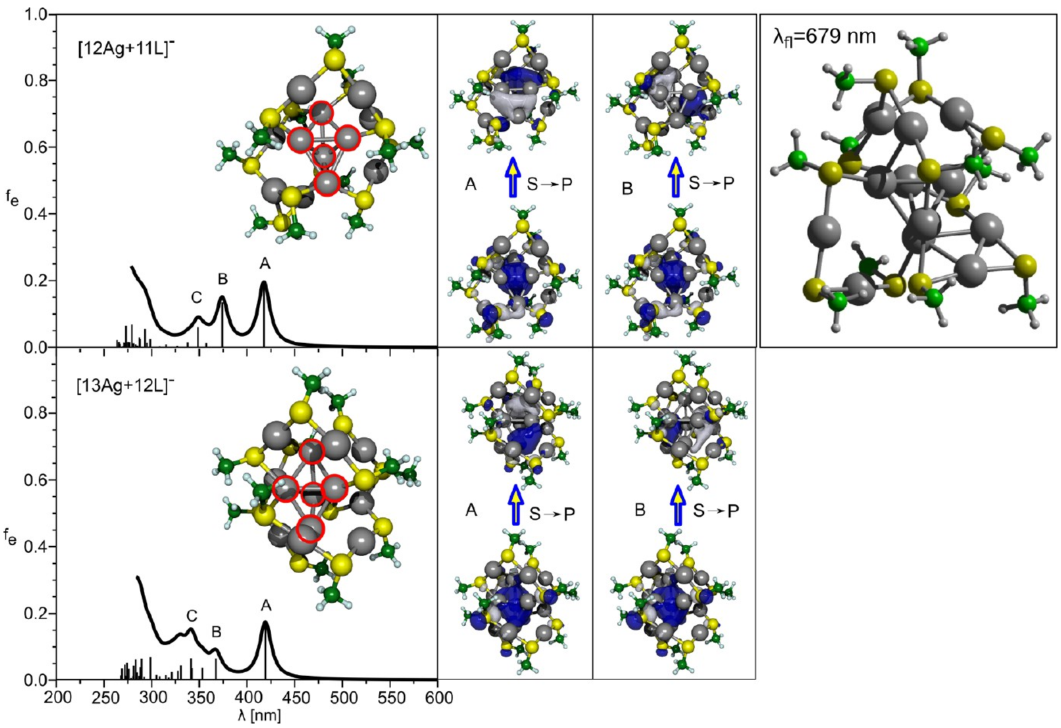

Figure 5. (Left) Calculated absorption spectra for $[n\text{Ag}+(n-1)\text{L}]^-$ complexes (n = 12, 13; L = SCH_3) with two confined electrons. The spectra have been broadened by a Lorentzian width of 15 nm. (Middle) Leading excitations and involved orbitals in transitions A, B, and C arising between S and P orbitals (only two of three P components are shown) of the silver core (red circles) containing five silver atoms. (Right) Structure of $[12\text{Ag}+11\text{L}]^-$ optimized in the first excited state as well as the position of the fluorescence line.

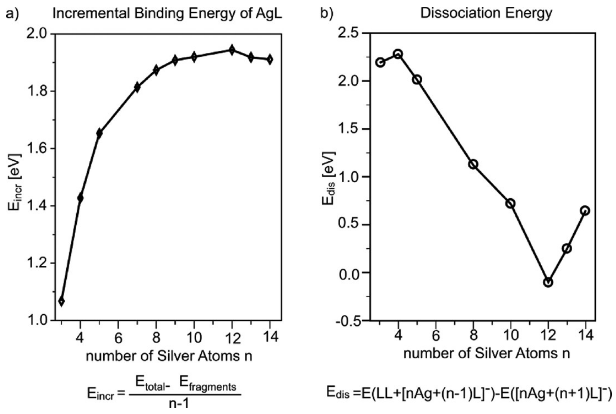

Figure 6. (a) Incremental binding energy of [Ag+SCH_3] subunits to [2Ag+SCH_3]$^-$. (b) Dissociation energy for the loss of [SCH_3−SCH_3] from systems with zero confined electrons forming thiolate-protected silver clusters with two confined electrons.

negative; thus, it would be energetically possible to generate the species with two confined electrons if the corresponding system with zero confined electrons is available. Therefore, the [12Ag+11L]$^-$ complex containing a fully ligand-protected Ag_5 core with two confined electrons, which exhibits intense absorption in the visible regime, might be an interesting candidate for experimental realization.

Complexes with four Confined Electrons. Anionic ligand-protected silver clusters with n silver atoms and $x = n - 3$ ligands contain four confined electrons. We present here results on three prototype systems with n = 4, 8, and 10 silver atoms. The first system that we considered contains the Ag_4 cluster subunit to which only one SCH_3 group is attached (see the inset in Figure 8). In contrast, in the case of eight Ag atoms and five SCH_3 ligands, the Ag_5 core is formed, in which four silver atoms are ligand-protected, as shown in Figure 7. For

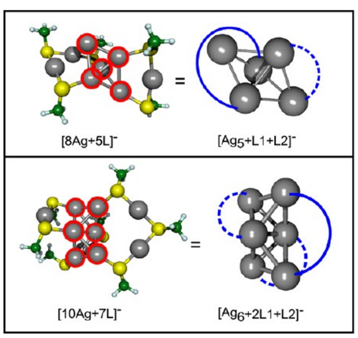

Figure 7. Structures of [nAg+(n−3)L]$^-$ (n = 8, 10) complexes with four confined electrons illustrating formation of a core with five and six silver atoms, respectively.

structures of higher-energy isomers, compare Figure S4 (Supporting Information). By increasing the number of silver atoms and ligands by two, the formation of a fully protected Ag_6 core in the [10Ag+7SCH_3]$^-$ complex can be achieved, as also shown in Figure 7. For all three above-described prototype systems, the calculated absorption spectra are presented in Figure 8. For the [4Ag+1SCH_3]$^-$ complex, the lowest-energy intense transition located at about 550 nm is characterized by the P → D type of excitation within the Ag_4 subunit. The spectra of the other two complexes with Ag_5 and Ag_6 cores exhibit the lowest-energy intense transitions at around 450 nm,

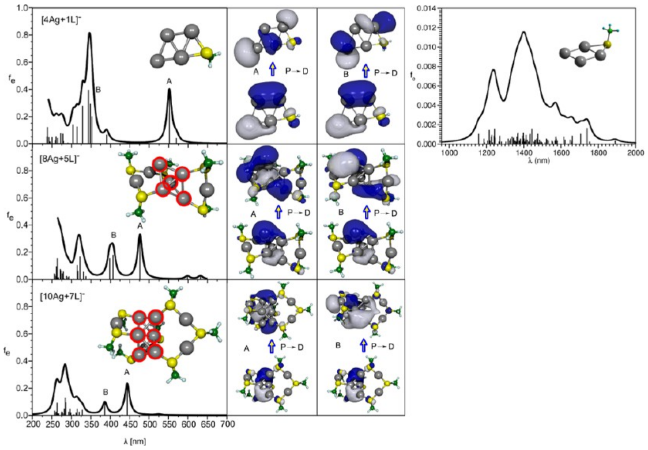

Figure 8. (Left) Calculated absorption spectra for [nAg+(n−3)L]$^-$ complexes (n = 4, 8, 10; L = SCH_3) with four confined electrons. The spectra have been broadened by a Lorentzian width of 15 nm. (Middle) Leading excitations and involved orbitals in transitions A and B arising between P and D orbitals of the silver core (red circles) containing five and six silver atoms. (Right) Broadened fluorescence spectra obtained from 100 structures sampled from a 0 K harmonic Wigner distribution around the optimized geometry of the S_1 state.

which are also due to P → D type excitations within the cluster core. Thus, we can conclude that the presence of low-energy intense transitions at around 400 nm and above is characteristic for systems with confined electrons. The calculated fluorescence spectrum for the smallest system with four confined electrons, shown on the right side of Figure 8, is strongly red shifted with respect to the corresponding absorption transition, giving rise to the fluorescence maximum at 1400 nm.

In order to investigate the confinement of electrons within the cluster core, we present in Figure 9 the analysis of the ELF in systems with cores of 2−5 Ag atoms. It can be seen that in the case of Ag_2 and Ag_3 cores, the confined electrons are

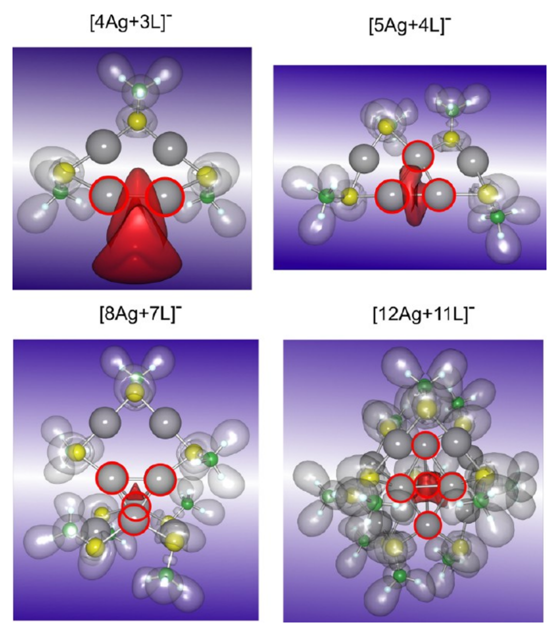

Figure 9. ELF analysis showing in red the spatial region in which the two confined electrons in the cluster core of [nAg+(n−1)L]$^-$ (n = 4, 5, 8, 12) complexes are localized.

located between two silver atoms, while in the case of the larger cores, the electrons are confined in the core cavity. In addition, the comparison of the ELF for the prototype systems without confined electrons with those containing two and four confined electrons, shown in Figure 10, illustrates the transition from systems with directional bonding (e.g., $[4Ag+5L]^-$) to systems with confined electrons within the cluster core (e.g., $[4Ag+3L]^-$ and $[4Ag+1L]^-$), which are responsible for intense optical absorption and emission.

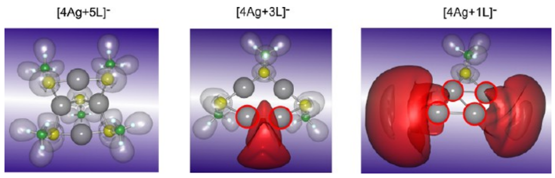

Figure 10. ELF analysis of liganded systems containing four silver atoms that exhibit either no cluster core or a core of two and four silver atoms with the corresponding number of confined electrons, respectively. The spatial region in which the confined electrons are localized is shown in red.

■ CONCLUSION

We have systematically investigated structural and optical properties of thiolate-protected silver clusters with the general composition $[nAg+xL]^-$, where $x = n + 1$, $n - 1$, and $n - 3$, which formally contain zero, two, and four confined electrons in the cluster core. Our findings illustrate that the number of confined electrons within the cluster cores in these sytems determines the spectroscopic patterns. For systems with two or four confined electrons, strong absorption and fluorescence in the visible or even IR energy regime can be found. Therefore, new stable fully ligand-protected clusters with tailored optical properties can be designed. Such species are highly interesting as potential building blocks for sensing and photonic materials.

■ ASSOCIATED CONTENT

Ⓢ Supporting Information

Figures S1 and S2 show structures and relative energies of low-lying isomers of systems with zero or two confined electrons, respectively. Figure S3 presents the comparison between the absorption spectra of $[5Ag+4L]^-$ with those of the SCH_3 and the SCH_2COOH ligands. Figure S4 shows structures and relative energies of low-lying isomers of systems with four confined electrons. This material is available free of charge via the Internet at http://pubs.acs.org.

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Notes

The authors declare no competing financial interest.

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